Dataset 1 — TXT (Website Login Logs)

File: **logins.txt**

```
Rahul
Sneha
Arjun
Rahul
Priya
Sneha
Rahul
Karan
Arjun
Sneha
Rahul
Amit
Priya
Karan
Sneha
Rahul
Meera
Arjun
Sneha
Rahul
Karan
Amit
Priya
Sneha
Rahul
Arjun
```

Questions (TXT Processing)

1. Read the TXT file and print all names.
2. Count the **total login events**.
3. Find the **unique users**.
4. Count how many times each user logged in.

5. Find the **top 3 most active users**.
6. Find users who logged in **more than 4 times**.
7. Convert the login list into a **dictionary of counts**.

Expected format example:

```
{
"Rahul": 7,
"Sneha": 6,
"Arjun": 4
}
```

Dataset 2 — CSV (Employees Dataset)

File: **employees.csv**

```
emp_id,name,department,salary,city
1,Rahul,IT,70000,Hyderabad
2,Sneha,HR,60000,Bangalore
3,Arjun,IT,75000,Chennai
4,Priya,Finance,80000,Hyderabad
5,Karan,IT,50000,Mumbai
6,Amit,HR,58000,Delhi
7,Meera,Finance,82000,Bangalore
8,Ravi,IT,72000,Hyderabad
9,Neha,HR,61000,Chennai
10,Vikram,Finance,90000,Delhi
11,Anita,IT,65000,Bangalore
12,Manoj,HR,62000,Mumbai
13,Divya,IT,77000,Hyderabad
14,Sanjay,Finance,85000,Chennai
15,Pooja,IT,69000,Bangalore
16,Kunal,HR,61000,Delhi
17,Sonal,Finance,88000,Mumbai
18,Deepak,IT,73000,Hyderabad
19,Ritu,HR,60000,Chennai
20,Akash,Finance,91000,Delhi
```

Questions (CSV Processing)

1. Load the CSV file.
2. Count total employees.
3. Show employees from **IT department**.
4. Find employees with salary **greater than 75,000**.
5. Calculate **average salary**.
6. Find **highest paid employee**.
7. Find **lowest paid employee**.
8. Count employees per department.
9. Calculate **average salary per department**.
10. Find how many employees are in each **city**.
11. Find the **top 5 highest salaries**.
12. Find employees working in **Hyderabad with salary > 70k**.

Dataset 3 — CSV (Sales Dataset)

File: **sales.csv**

```
sale_id,emp_id,product,quantity,price
1,1,Laptop,1,75000
2,2,Mouse,3,500
3,3,Keyboard,2,1500
4,1,Monitor,1,12000
5,4,Laptop,1,75000
6,3,Mouse,2,500
7,5,Keyboard,1,1500
8,1,Laptop,1,75000
9,6,Mouse,4,500
10,7,Monitor,2,12000
11,8,Keyboard,2,1500
12,9,Mouse,3,500
13,10,Laptop,1,75000
14,11,Monitor,1,12000
15,12,Keyboard,2,1500
16,13,Laptop,1,75000
17,14,Mouse,3,500
18,15,Monitor,1,12000
19,16,Keyboard,1,1500
20,17,Laptop,1,75000
```

Questions (Sales Analysis)

1. Calculate **revenue per sale**.
2. Calculate **total revenue**.
3. Find **revenue per product**.
4. Find **total quantity sold per product**.
5. Find **best selling product**.
6. Find **employee generating highest revenue**.
7. Find **average sale value**.
8. Find products generating **revenue above 100,000**.

Expected example output:

```
Laptop → 450000
Mouse → 7500
Keyboard → 15000
Monitor → 48000
```

Dataset 4 — JSON (Orders Dataset)

File: **orders.json**


```
{
  "orders": [
    {"order_id":1,"customer":"Rahul","product":"Laptop","amount":75000,"c
    {"order_id":2,"customer":"Sneha","product":"Mouse","amount":1500,"cit
    {"order_id":3,"customer":"Arjun","product":"Keyboard","amount":3000,"
    {"order_id":4,"customer":"Priya","product":"Laptop","amount":75000,"c
    {"order_id":5,"customer":"Karan","product":"Monitor","amount":12000,"
    {"order_id":6,"customer":"Rahul","product":"Mouse","amount":1000,"cit
    {"order_id":7,"customer":"Sneha","product":"Laptop","amount":75000,"c
    {"order_id":8,"customer":"Arjun","product":"Keyboard","amount":3000,"
    {"order_id":9,"customer":"Priya","product":"Mouse","amount":2000,"cit
    {"order_id":10,"customer":"Rahul","product":"Monitor","amount":12000,
  ]
}
```

Questions (JSON Processing)

1. Load the JSON file.
2. Print all orders.
3. Count total orders.
4. Calculate **total sales amount**.
5. Find **total spending per customer**.
6. Find **highest spending customer**.
7. Find **total sales per product**.
8. Find customers from **Hyderabad**.
9. Find orders with amount **greater than 10,000**.
10. Count how many orders were placed in each **city**.

Final Combined Challenge

Using all datasets:

Tasks

1. Join **employees.csv and sales.csv** using `emp_id`.
2. Find **total revenue generated by each employee**.
3. Find **top 5 employees by sales**.
4. Find **department generating highest revenue**.
5. Save final result to:

```
final_sales_report.csv
```

Example output:

```
Employee Revenue

Rahul → 162000
Priya → 75000
Arjun → 4500
```
